HOUSAM KAK

Product lead and software engineer for enterprise AI | Dubai, UAE

housamkak.com | blog.housamkak.com | blockforge.housamkak.com | github.com/HousamKak | housam.kak20@gmail.com

PROFILE

Product lead and software engineer delivering enterprise AI and financial technology systems for investment firms. Combines requirements and stakeholder work with architecture and hands-on development. Independently builds his own systems: an AI agent workspace (H), a programmable 3D world engine (BlockForge), offline-first retail software and AI developer tools. Mechanical engineer and mathematician by training.

EXPERIENCE

A global investment firm 2024 - present

Product Lead, AI Systems Pilot

Enterprise conversational AI pilot | Client assignment through OmniVista Solutions

- Head the pilot AI systems team, owning product direction and senior software delivery for a conversational platform that connects staff to company data.
- The platform is used by 200+ internal users to find company information and support operational workflows.
- Connect enterprise data through MCP integrations, Gremlin-based graph access, Azure Cosmos DB and Azure AI Search, running on distributed microservices.

OmniVista Solutions Apr 2023 - present

Consultant

Financial technology consultancy, Dubai | Selected client engagements

- **Healthcare investment firm:** led requirements and authored the BRD/TRD for an analyst recommendation portal; designed its architecture and trade submission to the Charles River order management system, and coordinated planning and CI/CD. Contributed to a reported 65% shorter trade-idea turnaround.
- **Global investment firm, data integration:** built C#/ASP.NET Core Windows services for ESG risk and market-data retrieval, released through Bitbucket, TeamCity and Octopus Deploy; traced data and optimized queries, cutting portfolio application load time by about 70%; designed Power BI reporting for Expected and Target Exposure.
- **Hedge fund client:** led client support operations across 2,500+ tickets, automating support-data retrieval and analysis with Python and SQL.
- **Wealth management firm:** automated Excel/VBA dashboards and SQL extraction and import preparation for trade, security, position and fund data.

American University of Beirut, Interdisciplinary Design Practice Program Jun - Aug 2021

Engineering Intern

- Designed and manufactured a bottle sticker-placement mechanism in SolidWorks using AUB manufacturing facilities, improving placement accuracy over manual application.

SELECTED SYSTEMS

Independent work, 2026. Case studies at housamkak.com/work.

H | AI agent workspace housamkak.com/work/h-system

TypeScript, Node.js, React, Tauri, SQLite, MCP

An event-driven canvas where AI agents, terminals, project knowledge and live applets are wired together. Designed and built solo: an orchestration core for agents, tasks, sessions and MCP tools, with web, desktop and Telegram front ends.

BlockForge | Programmable 3D world engine housamkak.com/work/blockforge | blockforge.housamkak.com

C++17, SDL2, JavaScript, Three.js, Luau/WASM

Browser and native clients share schema-driven world formats and a Luau mod host, with an object compiler, world versioning and CLI/MCP control so agents can build, inspect and film scenes. Built end to end; the public site is rendered from the engine.

Store POS ecosystem | Offline-first retail software housamkak.com/work/pos-store-management

Node.js, Express, SQLite, Turso, Cloudflare R2, PostgreSQL

Keeps a Lebanese store selling without internet: transactional checkout, FIFO lot inventory, USD/LBP books, audit trails and outbox replication to the cloud, plus photo-based product discovery. Lead author, with contributions from Mahdi Kak.

AnyThyng | Sentence to working phone app housamkak.com/work/anythyng | anythyng.app

TypeScript, Claude Agent SDK, Kotlin/Compose, AWS, OpenTofu

A server that generates, versions and sandboxes apps from a sentence, an Android shell with a native bridge, and OpenTofu AWS infrastructure with OIDC release pipelines.

UIpen | Point-and-fix UI tool for coding agents housamkak.com/work/uipen | uipen.dev

TypeScript, React/Next.js, Bun, MCP

A dev-mode overlay for pinning problems on a running app; each annotation carries the selector, source file, line and component into a Claude Code session, so the agent edits the right code. Built solo, with installers, licensing and the uipen.dev site.

Also: H Distribution (multi-platform publishing), Studio (script-to-film pipeline), Flying Screen (simulation and research paper), and co-founder and CTO of Atlagene, a genomics venture. Full index at housamkak.com/work.

EDUCATION

American University of Beirut 2018 - 2022

B.E. Mechanical Engineering, major in Mathematics (AUB)

Graduation project: autonomous miniature racing car with camera-based lane perception and pure-pursuit trajectory control.

RECOGNITION AND LANGUAGES

- Mastercard Foundation Scholar: full undergraduate scholarship at the American University of Beirut.
- Zari3tech: co-founded a social venture and secured $5,000 in funding through the 2021 Resolution Social Venture Challenge.

Languages: Arabic (native), English (full professional proficiency)

SKILLS

AI and architecture: Conversational AI, Model Context Protocol (MCP), agent orchestration, Gremlin, Azure Cosmos DB, Azure AI Search, distributed microservices

Engineering: TypeScript, JavaScript, Python, C#, C++, SQL, Kotlin; Node.js, React, Next.js, ASP.NET Core, Three.js

Data: PostgreSQL, SQLite, SQL Server, MySQL, Turso

Cloud and DevOps: Azure, AWS, Cloudflare, OpenTofu, Docker, CI/CD, TeamCity, Octopus Deploy, Git

Product and consulting: Requirements elicitation, BRD/TRD, stakeholder management, technical planning, Power BI, Excel/VBA